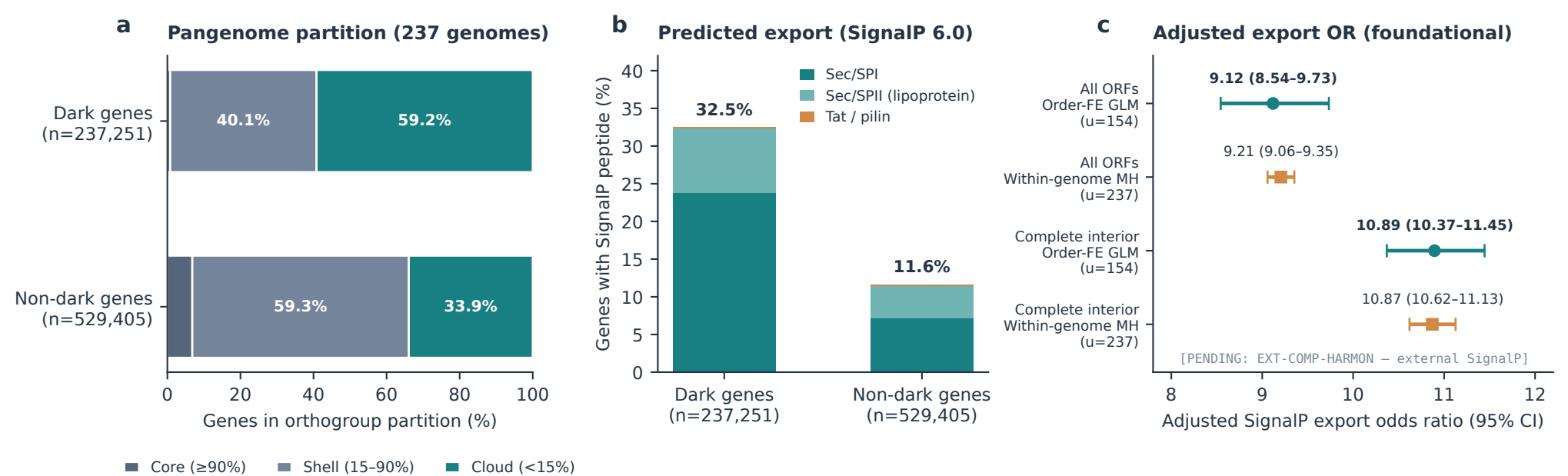
a
Pangenome partition (237 genomes)
Dark genes
(n=237,251)
40.1%
59.2%
Non-dark genes
(n=529,405)
59.3%
33.9%
0
20
40
60
80
100
Genes in orthogroup partition (%)
Core (≥90%)
Shell (15–90%)
Cloud (<15%)
b
Predicted export (SignalP 6.0)
Genes with SignalP peptide (%)
0
5
10
15
20
25
30
35
40
Sec/SPI
Sec/SPII (lipoprotein)
Tat / pilin
32.5%
11.6%
Dark genes
(n=237,251)
Non-dark genes
(n=529,405)
c
Adjusted export OR (foundational)
All ORFs
Order-FE GLM
(u=154)
9.12 (8.54–9.73)
All ORFs
Within-genome MH
(u=237)
9.21 (9.06–9.35)
Complete interior
Order-FE GLM
(u=154)
10.89 (10.37–11.45)
Complete interior
Within-genome MH
(u=237)
10.87 (10.62–11.13)
[PENDING: EXT-COMP-HARMON — external SignalP]
8
9
10
11
12
Adjusted SignalP export odds ratio (95% CI)